

REVIEW ARTICLE

# Antibiotic resistance crisis

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## ABSTRACT

Antimicrobial resistance (AMR) is a universal public health problem affecting the benefit of antibiotics. Antibiotic use, inappropriate dosage, and genetics led to the spread of multidrug resistance among pathogenic bacteria, and it became hard to treat infections. Here, we reviewed recent updates of antimicrobial resistance in the period from 2004 to 2019. Scientific websites were used to search for articles related to our subjects such as PubMed and Google scholar using several keywords including AMR, antibiotic abuse, antibiotic resistance, crisis, and bacterial infection. We obtained 55 articles, 45 of them published between the years 2004 and 2019 were included, whereas those that were excluded were out the scope of current review. Antibiotic resistance crisis is widespread and still increasing, due to continuous abuse of antibiotics and deficiency of the new antibiotic industry. A multifaceted plane is required to control the antibiotic resistance crisis. Educational programs and awareness campaigns are required. Research, explanations, and solutions should be implanted. Screening and diagnostic tools of antibiotic resistance should be improved and globally applied. Overall, AMR is not uncontrollable anymore. Antibiotics alternatives as probiotics and lytic bacteriophages can participate in decreasing AMR worldwide.

**Keywords:** AMR, antibiotic abuse, antibiotic resistance, crisis, bacterial infection.

## Introduction

In the early 19th century, infectious diseases were the major cause of death worldwide. The selective action exerted on pathogenic bacteria by the antibiotic created an important era for human beings ever since [1,2]. However, microorganism's resistance to antibiotics threatens antibiotic effectiveness, as well as the use of antibiotic incites resistance [3]. Antibiotic is a low-molecular weight compound, which is mostly produced by microorganisms and is active at reduced concentrations against other microorganisms. For example, Fluoroquinolones are natural products derivative (the bark of the Cincona tree), while sulfa and oxazolidinones do not originate from natural products [3,4].

Antibiotic resistance is a global health Chrysler, decreasing the efficacy of antibiotics, which keep millions of patients alive [5–11]. Again, bacterial infections become a life threat [12]. Antibiotic resistance crisis is strongly related to the overuse and abuse of antibiotics, and lack of the new antibiotic industry. Coordinated efforts are required to manage such crisis [13,14].

## Materials and Methods

The present review included 55 research articles that were obtained through online research. We used several keywords to search for articles related to our subjects in scientific websites such as PubMed and Google scholar,

these keywords included: AMR, antibiotic abuse, antibiotic resistance, crisis, and bacterial infection. We obtained 55 articles, 45 of them included and 10 were excluded as they involved other related topics and did not focus on our aim. The included articles published between the years 2004 and 2019.

## Discussion

Antibiotics are one of the major innovations of medicine. They can decrease the patient's life span by improving the outcome of several bacterial infections. Moreover, antibiotics decrease the morbidity and mortality burden by food-borne and surgical infections [15].

Among all antimicrobial types, antibiotics lose their effectiveness over time due to the development of antibiotic-resistant strains among bacterial pathogens. The time between the development of antibiotic resistance

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and the administration of antibiotic is been variable due to the dynamics of resistance dissemination [16].

In regards to the mechanism of antibiotic resistance development, horizontal gene transfer (HGT) is the main mechanism by which bacteria acquire antibiotic resistance. Bacterial cells can exchange genetic material. Antibiotic resistance genes are expressed by plasmids, transposons, or integrons which act as vectors transferring genetic data among bacteria and even between species. Chromosomal mutation is another mechanism of antibiotic resistance. The main difference between HGT and chromosomal mutation is resistance adaptation time occurrence in bacteria [17].

Minimum inhibitory concentration of antibiotic is the technique that uses the minimum dose that inhibits bacterial growth [18]. However, antibiotics abuse led to the appearance of bacteria that can withstand the inhibiting concentrations, and the bacteria persist [19].

Anthropogenic antibiotic use spreads resistance in the environment. Therapeutic antibiotic use and development of human antibiotic resistance are well known [20,21]. Anthropogenic activity elevates antibiotic concentrations in the environment due to excreted antibiotics [22], the animal feed containing the antibiotic is not all eaten, and the spillover of the antibiotic production industry [23].

Many types of antibiotics resistance resulted in strains with multidrug-resistance (MDR) phenotypes, and treatment options for those pathogens became limited. The high-risk MDR clones can cause a rapid epidemic diffusion due to cross infection and spread [24].

Extremely drug-resistant (XDR) or totally drug-resistant (TDR) phenotypes occurred when Gram-negative bacteria resistance include almost all the available antibiotics [25].

Antibiotic resistance crisis is related to lag in antimicrobial discovery and development programs in recent years and has recently drawn the attention of scientific societies, public health agencies, and political bodies [26,27].

*Staphylococcus aureus* and *Enterococcus* spp. are the most common Gram-negative bacterial species which developed antibiotic resistance. Methicillin-resistant *S. aureus* (MRSA), is a universal cause of Gram-positive antibiotic resistance crisis [28,29]. MRSA phenotype is developed due to modified penicillin-binding proteins expression (PBPs), which encoded by acquired genes that alter the job of the resident staphylococcal PBPs and are not affected by conventional  $\beta$ -lactams. MRSA is a leading cause of human infections in many countries [30,31], however; MRSA dissemination can be prevented by aggressive antimicrobial control [32].

Endemic MRSA means limited MRSA dissemination by effective infection control [33]. Some antimicrobials are active against MRSA, by controlling modified PBPs which are responsible for the methicillin-resistance [34]. Although MRSA is a manageable antibiotic crisis, it

emerges and spread in different epidemiological settings (hospital, community, and animals) over time [35,36]. Resistance to the anti-MRSA agent is by mutation, as well as transferable resistance to linezolid and glycopeptides was recorded with MRSA [37,38].

Vancomycin-resistant *Enterococci* (VRE) have a lower epidemiological effect than MRSA. Although VRE worldwide prevalence is relatively low (with the exception of the United States and some European countries) [39], it plays a major effect in the hematology-oncology patients as well as the ongoing antibiotic resistance crisis [40].

The Gram-negative antibiotic crisis is more serious than with the Gram-positives. XDR and TDR phenotypes among Gram-negative bacteria are more prevalent as *Pseudomonas aeruginosa*, *Acinetobacter* spp., and Enterobacteriaceae. Also, MDR Gram-negatives are prevalent including *Escherichia coli* and *Neisseria gonorrhoeae* [41,42].

Antibiotic-resistant infections are widespread worldwide. The 2011 national survey of infectious disease, done by the Infectious Diseases Society of America (IDSA) Emerging Infection Network, demonstrated that 60% of doctors in the study had seen a multi-resistant bacterial infection did not respond to treatment in the previous year [11].

Some public health organizations described the rapid onset of resistant bacteria as a “crisis” that led to catastrophic complications” [12].

In 2013, The Centers for Disease Control and Prevention (CDC) announced that the “post-antibiotic era” is started and the WHO in 2014 said that the antibiotic resistance crisis became dire [14].

*Staphylococcus aureus* and *Enterococcus* species (among Gram-negative resistance) created a worldwide pandemic infection, and responsible for a big threat [9,43], while *Streptococcus pneumonia* and *Mycobacterium tuberculosis* are responsible for epidemic Gram-negative resistant respiratory infections [43].

The most serious Gram-negative infections are hospital acquired and the common pathogen is Enterobacteriaceae (mostly *Klebsiella pneumonia*), *Pseudomonas aeruginosa*, and *Acinetobacter*. Bacterial resistance is a serious threat and demand monitoring and prevention activities [9,43].

Global antibiotic resistance does not seem to stop but it may change its direction. The cause of antibiotic resistance is multifactorial, and its complications affect the globe. A coordinated global campaign is required, in an attempt to control the antibiotic-resistant crisis [44].

The success of modernized medicine such as organ transplantation and advanced major surgeries is dependent on antibiotic effectiveness in the management of bacterial infections. So, the failure of this crisis control will lead to serious social, medical, and economic consequences [45].

## Conclusion

Antibiotic resistance crisis is widespread and still increasing, due to continuous abuse of cocaine and deficiency of the new antibiotic industry. A multifaceted plane is required to control the antibiotic resistance crisis. Educational programs and awareness campaigns are required. Research, explanations, and solutions should be implanted. Screening and diagnostic tools of antibiotic resistance should be improved and globally applied. Overall, AMR is not uncontrollable anymore. Antibiotics alternatives as probiotics and lytic bacteriophages can participate in decreasing AMR worldwide.

## List of Abbreviations

|      |                                        |
|------|----------------------------------------|
| AMR  | Antimicrobial resistance               |
| HGT  | Horizontal gene transfer               |
| MDR  | Multidrug-resistance                   |
| MRSA | Methicillin-resistant <i>S. aureus</i> |
| PBPs | Penicillin-binding proteins expression |
| TDR  | Totally drug-resistant                 |
| VRE  | Vancomycin-resistant enterococci       |
| XDR  | Extensively drug-resistant             |
| XDR  | Extremely drug-resistant               |

## Conflict of interest

The authors declare that there is no conflict of interest regarding the publication of this article.

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